

RSIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool: Technical Presentation (Low)

Assessment Tool Weightage: 10%

Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Judy Allen J

TECHNICAL PRESENTATION

CSA0509 – Database Management Systems
CO1 – Assessment Tool 3

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ER Diagram

Definition

- ▶ Graphical representation of a database.
- ▶ Shows Entities, Attributes, and Relationships.
- ▶ Used before database implementation.

Objectives

- ▶ Organise data
- ▶ Reduce redundancy
- ▶ Improve consistency
- ▶ Simplify database design

ER Diagram Components

Entity

- ▶ Real-world object
- ▶ Example: Member, Book

Attribute

- ▶ Describes an entity
- ▶ Example: Name, BookID

Relationship

- Connects entities
- Example: Borrows, Issues

Keys

- Primary Key (PK)
- Foreign Key (FK)

ER Diagram for an Online Library System

Entities

- ▶ Member
- ▶ Book
- ▶ Librarian

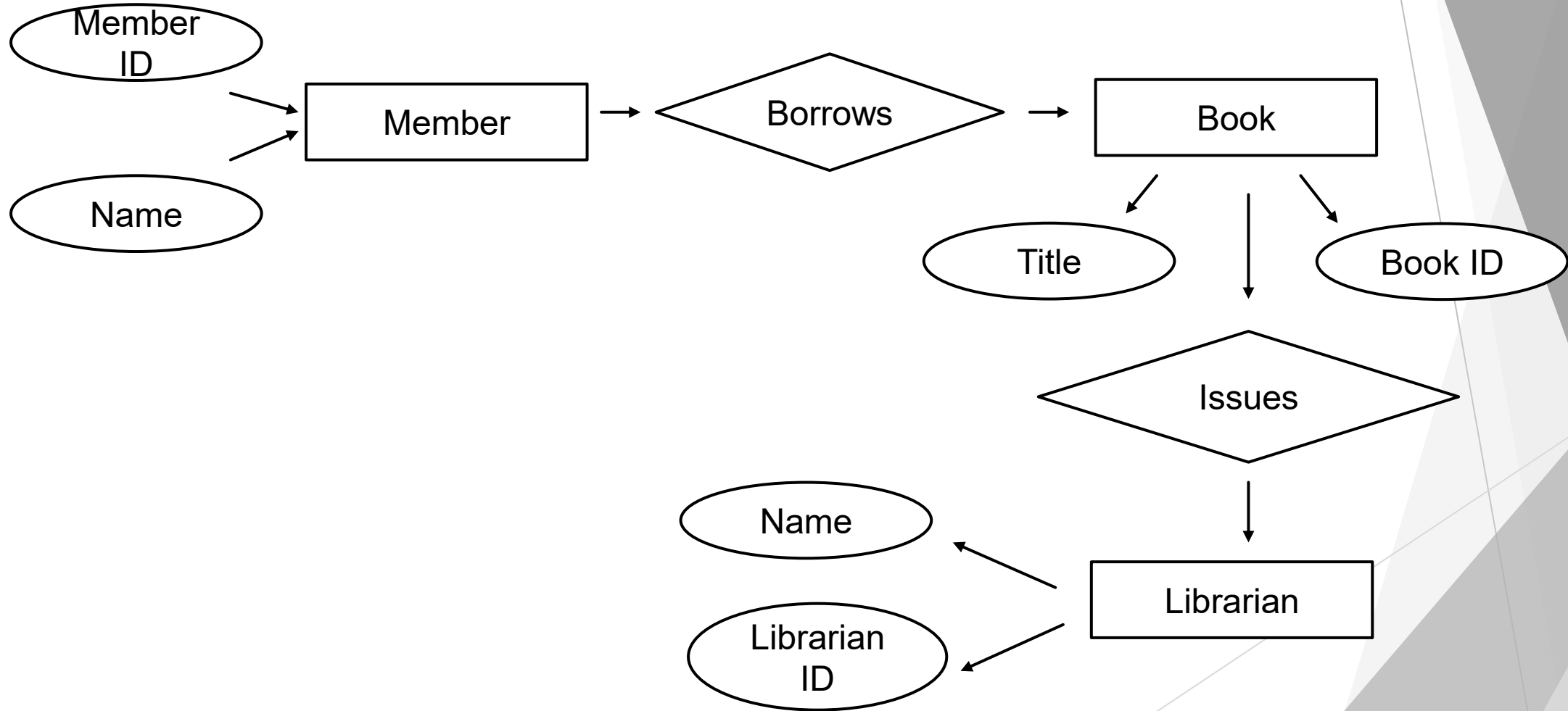
Relationships

- ▶ Member → Borrows → Book
- ▶ Librarian → Issues → Book

Advantages

- Easy database design
- Maintains integrity
- Reduces duplication
- Efficient data retrieval

ER Diagram for an Online Library System



Enhanced Entity Relationship (EER) Model

Definition

- ▶ Extension of the ER Model.
- ▶ Represents complex real-world data.
- ▶ Supports inheritance.

Advantages

- ▶ Better data modelling
- ▶ Reduces redundancy
- ▶ Improves database flexibility
- ▶ Supports inheritance

Generalization

Definition

- ▶ Combines similar entities into one
- ▶ Common attributes are stored in the superclass.

Benefits

- ▶ Eliminates duplicate data
- ▶ Improves consistency
- ▶ Easier maintenance.

University Example

Student
Faculty
Staff



Person (Superclass)

Specialization

Definition

- ▶ Divides one Superclass into multiple Subclasses.
- ▶ Each subclass has unique attributes.

Applications

- ▶ Student Management
- ▶ Faculty Records
- ▶ HR Management

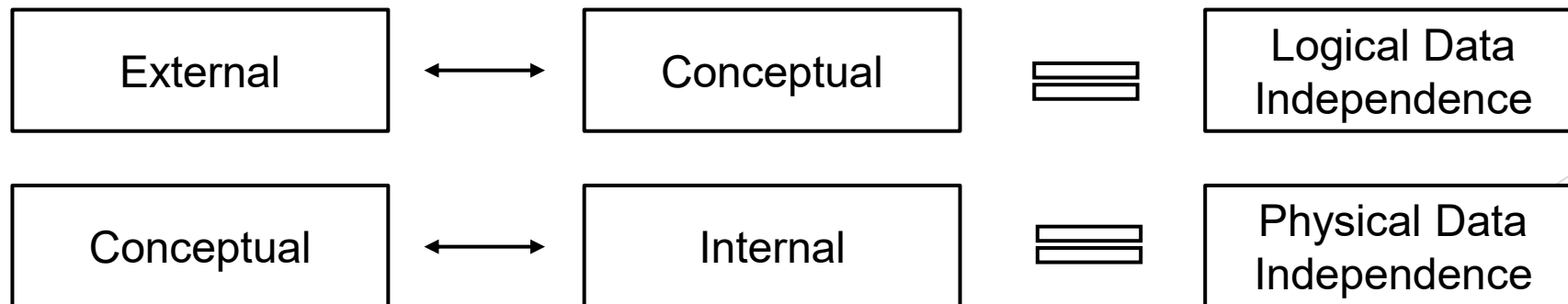
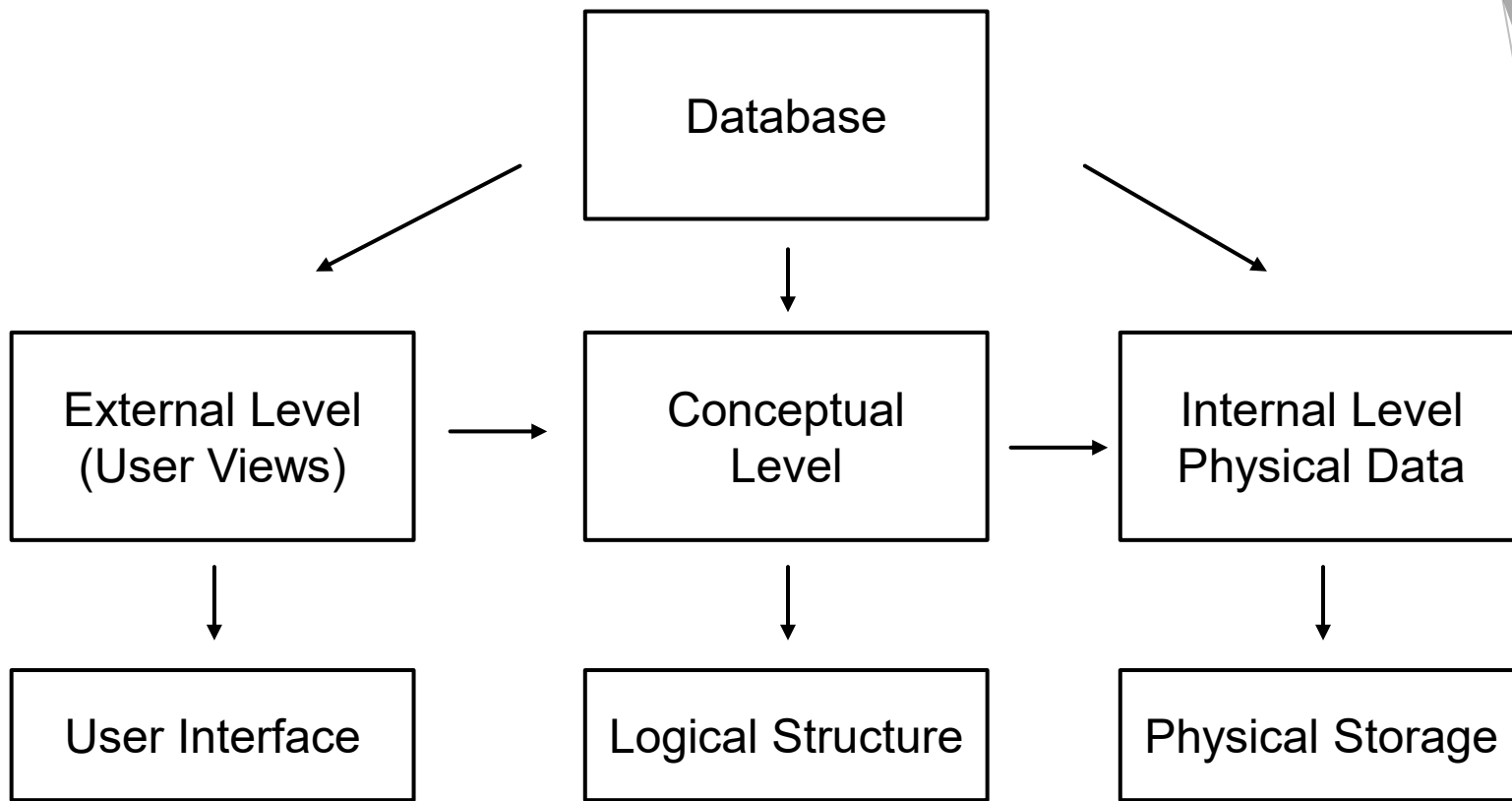
University Case Study Person

- Student
- Faculty
- Staff

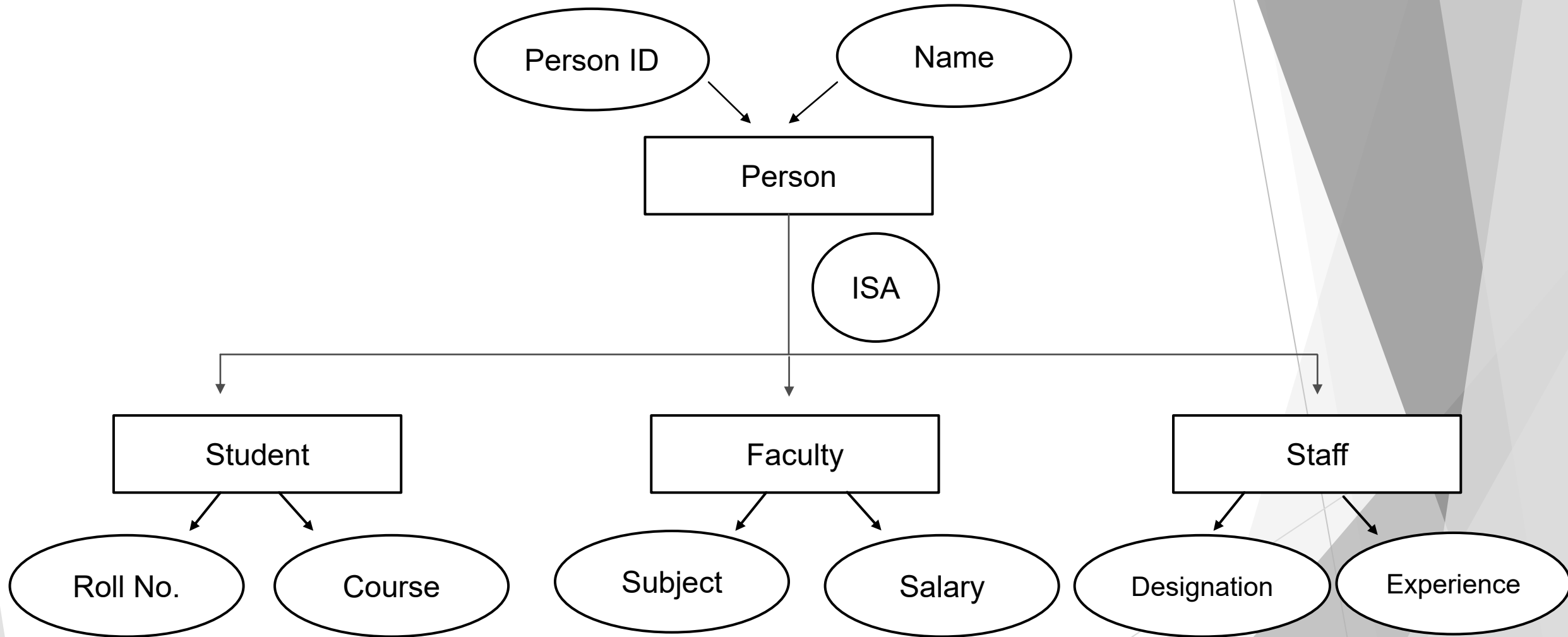
Student → Roll No.,
Course

Faculty → Salary, Subject

Staff → Designation



EER Diagram (University System)



Data Independence

Definition

- Ability to modify one level of the database without affecting another level.
- Ensures flexibility in database management.

Types

Logical Data Independence

Physical Data Independence

Importance

- Easy maintenance
- Better flexibility
- Supports database updates
- Reduces application changes

Types of Data Independence

Logical Data Independence

- Changes the Conceptual Schema.
- Does not affect user applications.

Example

Adding a new column to a table.

Physical Data Independence

- Changes the Internal Storage.
- Does not affect the logical structure.

Example

Changing file organisation or indexing.

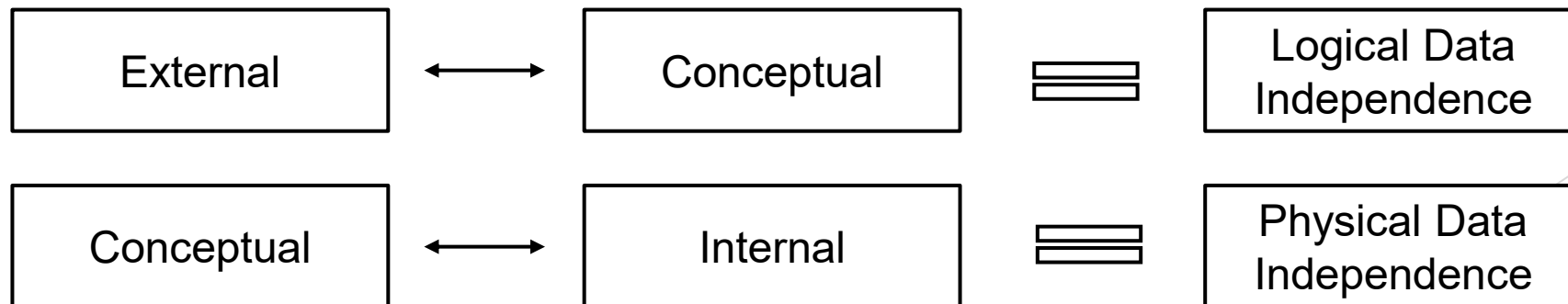
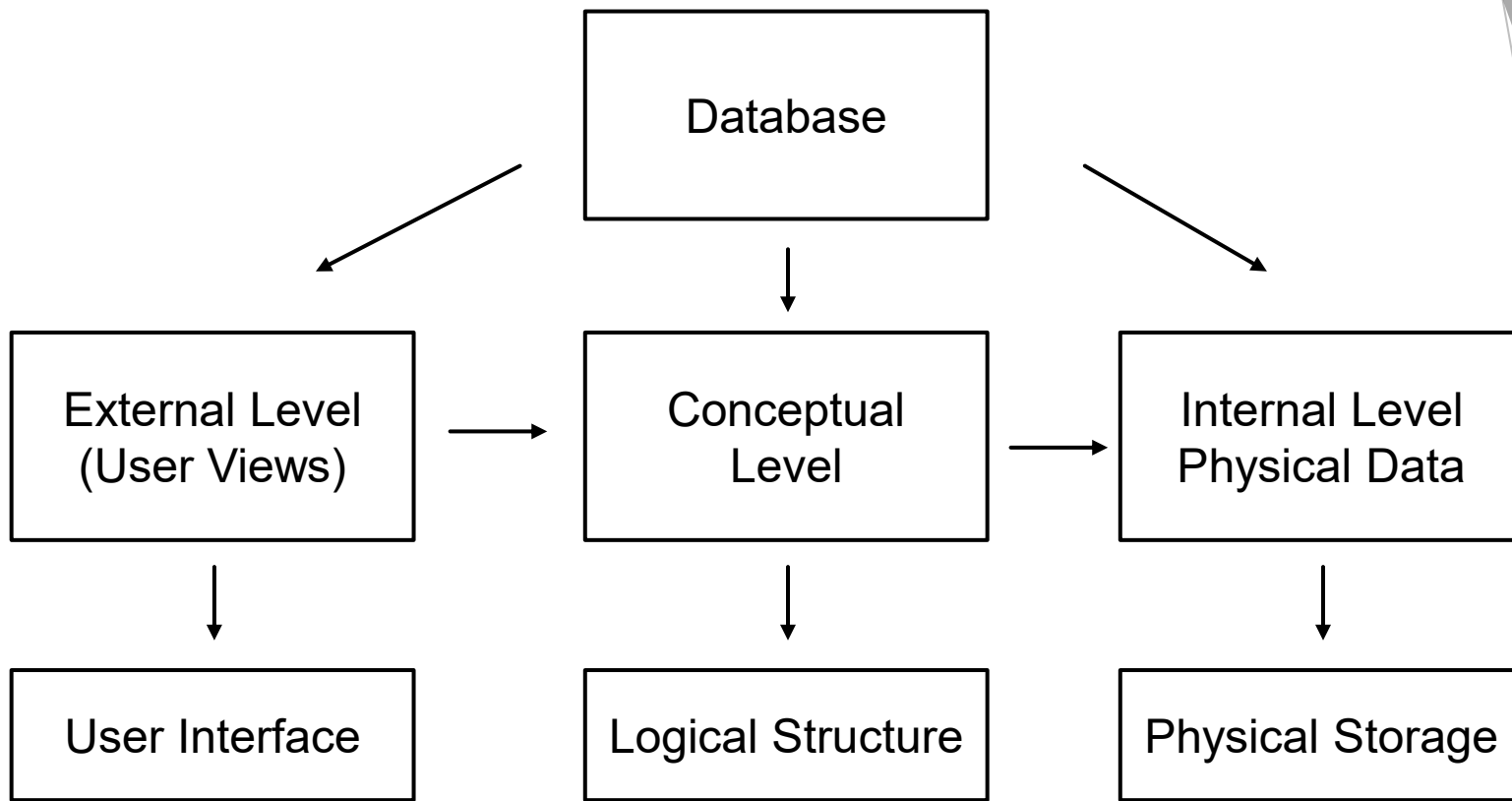
Significance in DBMS Architecture

Benefits

- Easy database modification
- Improved performance
- Better data security
- Reduced maintenance cost
- Supports scalability

Real-Life Applications

- Banking System
- Hospital Management
- Library Management
- University Database



Software Components of DBMS

Definition

- DBMS consists of software components that manage database operations.
- These components process queries, store data, and maintain transactions.

Main Components

- Query Processor
- Storage Manager
- Transaction Manager

Functions

- Query execution
- Data storage
- Transaction management
- Database security

Query Processor & Storage Manager

Query Processor

- Interprets SQL queries.
- Optimises query execution.
- Retrieves requested data.

Storage Manager

- Manages database files.
- Controls memory buffers.
- Organises physical storage.

Advantages

- Faster query execution
- Efficient data access
- Better storage management

Transaction Manager

Definition

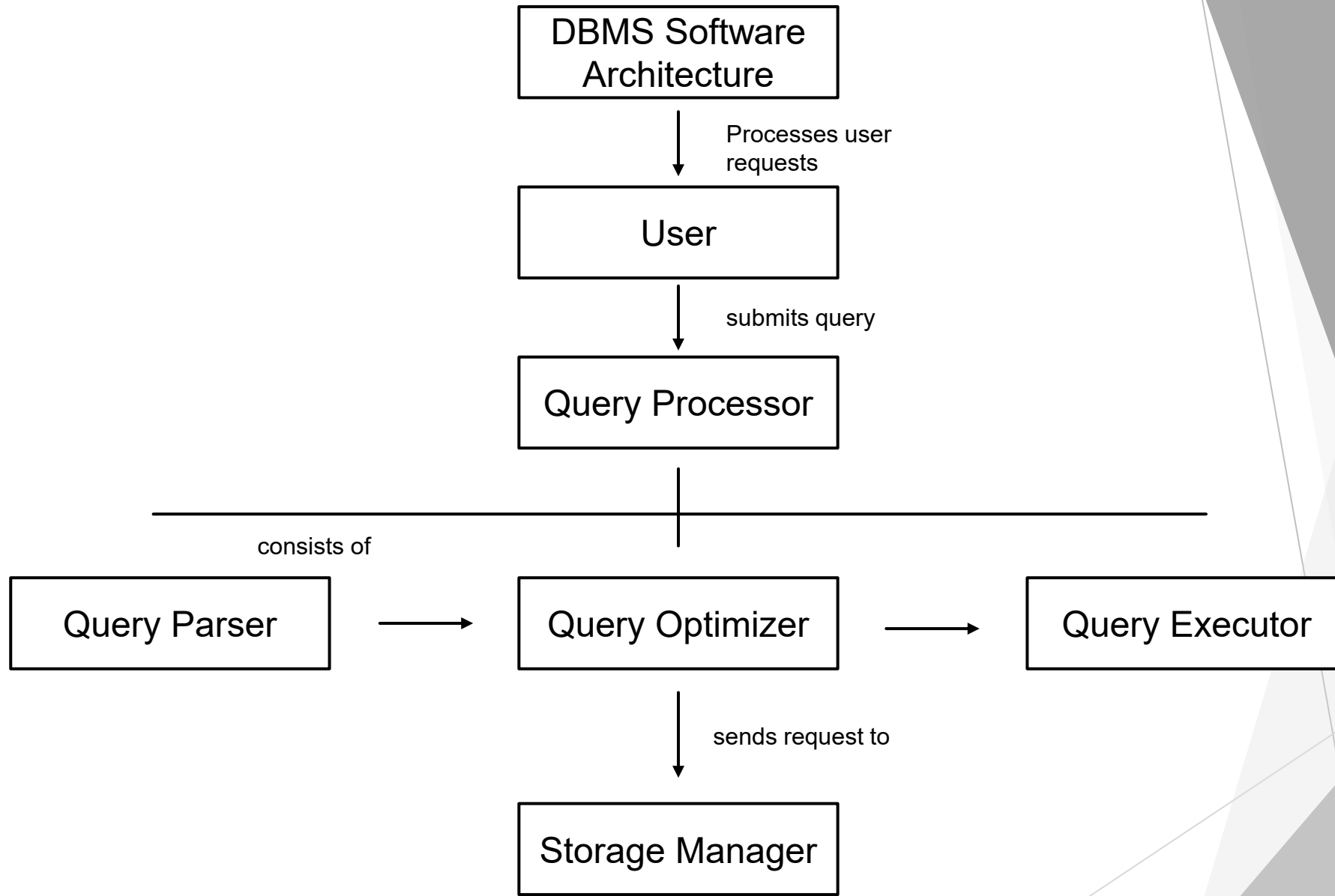
- Controls all database transactions.
- Ensures data consistency.
- Maintains ACID properties.

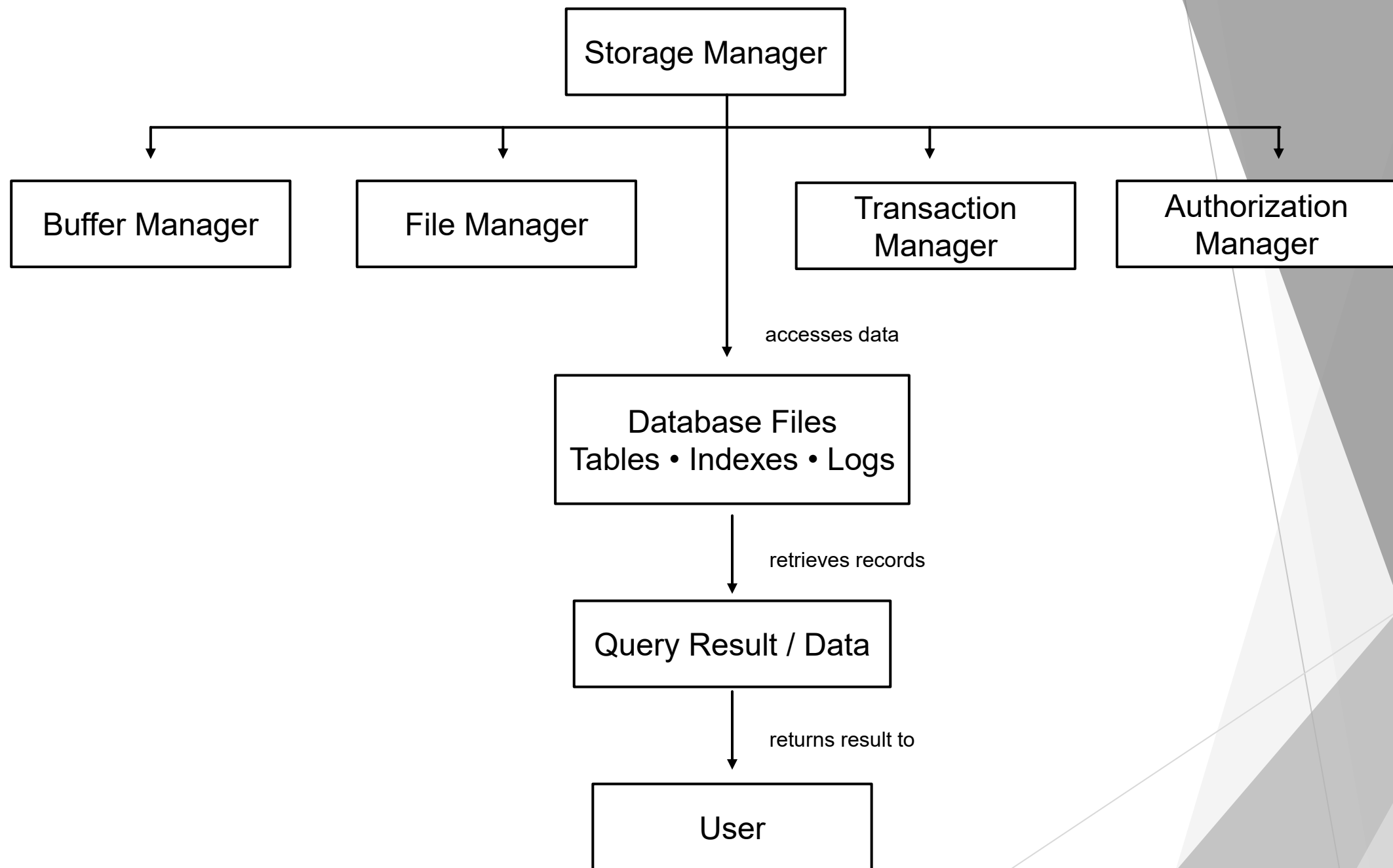
Responsibilities

- Concurrency Control
- Recovery Management
- Rollback & Commit
- Error Handling

Applications

- Banking System
- ATM Transactions
- Online Shopping
- Railway Reservation





Strong and Weak Entities

Definition

- An Entity represents a real-world object in a database.
- Entities are classified as:
 - Strong Entity
 - Weak Entity

Importance

- Identifies database objects.
- Maintains relationships.
- Improves database design.
- Ensures data integrity.

Strong Entity

Definition

- Has its own Primary Key.
- Exists independently.

Characteristics

- Independent entity
- Unique identification
- Single rectangle representation

Example Student

Student_ID (PK)

Name

Department

Weak Entity

Definition

- Does not have a Primary Key.
- Depends on a Strong Entity.

Characteristics

- Dependent entity
- Uses a foreign key
- Double rectangle representation

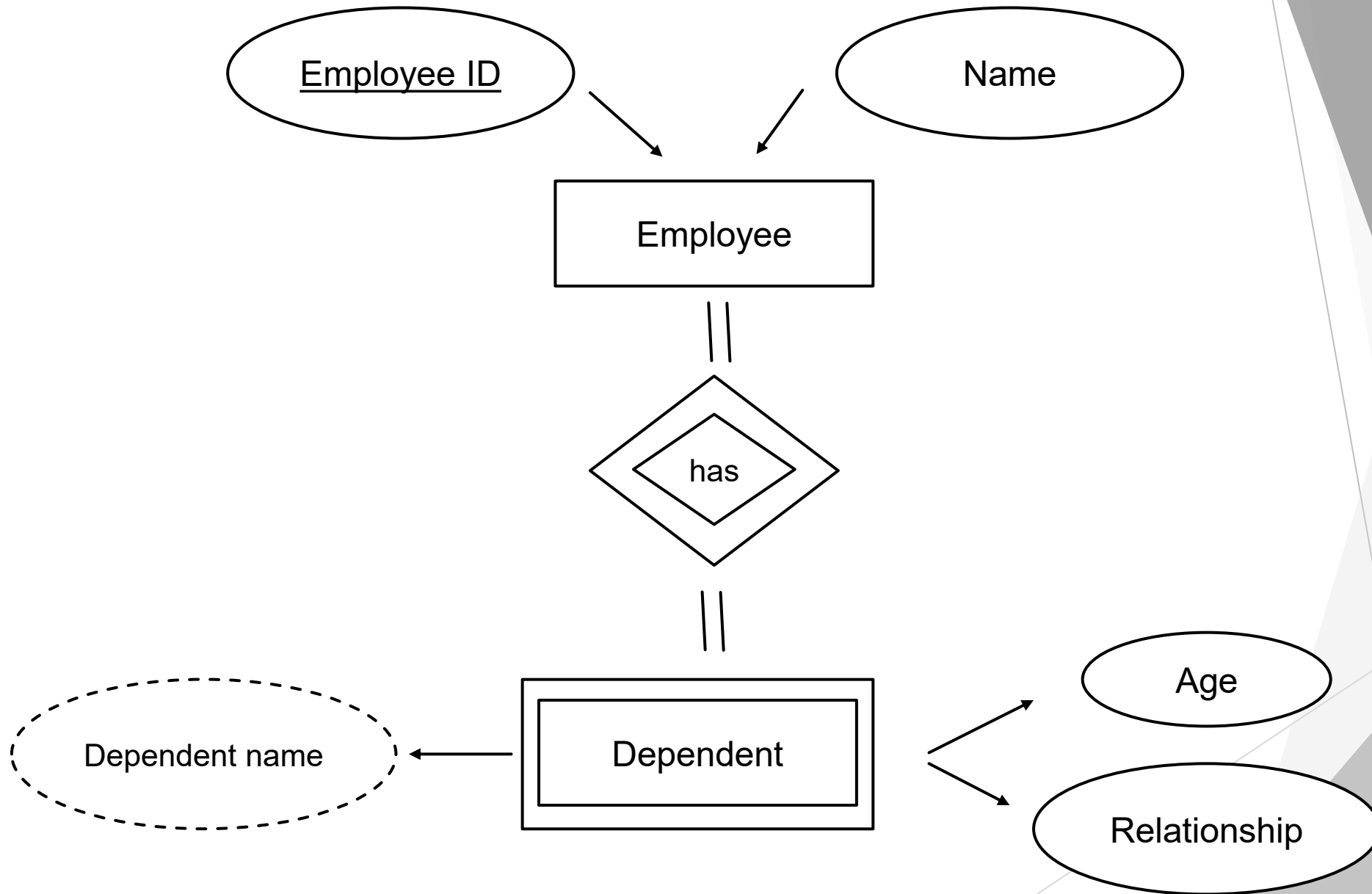
Example

Dependent

Dependent_Name

Age

Depends on Employee



THANK YOU